# A062 — SST Microgravity Force-Gradient Falsifier

v0.1.0 Scientific Protocol and Standalone Numerical Report

SST-Workbench — Candidate Model Analysis

12 October 2026

**Status: restricted model diagnostics, not a test of SST canon as a whole.** The target-bearing data of the published article are already public and are therefore *not* scientifically blind. The science-only run is separate from the canonical frozen v1.0.7 pipeline; no framework certification or real reveal is claimed. Protocol SHA-256 (prefix): `5dd2f0d5ce2df29b`.

## 1 Question, provenance and expected falsifiers

Can an explicit incompressible, inviscid, localized azimuthal Gaussian vortex yield both a mass-equivalent source and a gravitational far-field force gradient with $F \propto r^{-2}$, without injecting $G$ or fitting its amplitude? The mechanical instrument uses a 100-g test mass and a movable 1.2-g source at 4.2 cm separation, with 1.4 $\mu$m modulation. These are public rounded values reported by Zhu et al. [1]. The tested candidate is *not* a PKLSA knot and the surrogate source width is a chosen coarse-graining scale $a = 3\,\mathrm{mm}$, not the canonical $r_c$.

**Null and alternative.** Under H0 the restricted Gaussian mass/pressure closure fails at least one pre-registered scaling/mass test. Under H1 this closure passes both the mass ratio (within 10%) and the inverse-square slope $(2 \pm 0.2)$, on both source orientations, with zero fitted parameters. No general SST conclusion can be inferred from rejecting H1.

## 2 Comparator and mechanical readout

The Newtonian benchmark is

$$F_N = \frac{G m_s m_t}{r^2}, \qquad \left|\frac{dF_N}{dr}\right| = \frac{2G m_s m_t}{r^3}, \qquad \Delta F_N \simeq \left|\frac{dF_N}{dr}\right| \Delta r. \tag{1}$$

Use $G = 6.67430 \times 10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}$ as the *benchmark input only*. The published force values have a Newton-calibrated interpretation and cannot be used as an independent determination of $G$ or $F_{\mathrm{gr}}^{\max} = c^4/(4G)$.

The reproduced static force is $4.54034e-12\,\mathrm{N}$; the force gradient is $2.162067e-10\,\mathrm{N/m}$ and the first-order modulation is $3.026893e-16\,\mathrm{N}$. The reported rounded target is $3.02 \times 10^{-16}\,\mathrm{N}$. Apparatus response, spatial geometry and detector noise must be modelled independently before inference from raw traces.

## 3 Euler pressure source and dimensional audit

An exact divergence-free *initial velocity field* (not a claimed stationary Euler solution) is

$$\mathbf{v}(\mathbf{x}) = \Omega \exp\left(-\frac{x^2+y^2+z^2}{2a^2}\right)(-y, x, 0), \quad \Omega = \frac{e^{1/2}|\mathbf{v}_\circlearrowleft|}{a}. \tag{2}$$

Here $\rho_f = 7.0 \times 10^{-7}\,\mathrm{kg\,m^{-3}}$ and $|\mathbf{v}_\circlearrowleft| = 1.09384563 \times 10^{6}\,\mathrm{m\,s^{-1}}$. With $\nabla \cdot \mathbf{v} = 0$, the instantaneous Euler pressure-Poisson source is

$$\nabla^2 p = -\rho_f\, \partial_i v_j\, \partial_j v_i = 2\rho_f \Omega^2 e^{-r^2/a^2} \left(1 - \frac{x^2+y^2}{a^2}\right) \equiv S(\mathbf{x}). \tag{3}$$

The infinite-domain solution and test-volume approximation used by the code are

$$p(\mathbf{x}) = -\frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{S(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}\, d^3y, \quad \mathbf{F}_p = -V_t \nabla p, \qquad V_t = \frac{4\pi a^3}{3}. \tag{4}$$

This $V_t$ is a deliberately fixed test-volume proxy; it is *not* independently identified with the physical 100-g test mass. $[S] = \mathrm{Pa\,m^{-2}}$ and $[V_t \nabla p] = \mathrm{N}$.

The Gaussian field has zero integrated monopole $\int S\, d^3x = 0$ on the infinite domain. Thus its pressure field has no $1/r$ scalar monopole contribution, and a generic $1/r^2$ force is not expected at large separations. Both axial and equatorial orientations are tested to avoid orientation cherry-picking.

## 4 Explicit (candidate) mass closure

The proposed *restricted* energy-mass identification is

$$m_E = \frac{1}{2c^2} \rho_f \int_{\mathbb{R}^3} |\mathbf{v}|^2\, d^3x = \frac{\rho_f \Omega^2 \pi^{3/2} a^5}{2c^2}. \tag{5}$$

No source mass is used to calibrate $\Omega$: it is fully set by the canon's characteristic speed and the fixed $a$. The predicted equivalent mass is $1.90424e-18\,\mathrm{kg}$ and $m_E/m_s = 1.586867e-15$ (experimental source: $1.2 \times 10^{-3}\,\mathrm{kg}$). Disagreement refutes *this* postulated closure, not every possible vortex mass theory.

## 5 Numerics, precision and gate outcomes

The instantaneous pressure gradient is evaluated by finite-volume midpoint integration of the free-space Green function over a cube extending to $5.5a$ in every direction. Both field orientations are computed at six separations from 0.03 to 0.08 m, with grid sizes $40^3, 56^3$ and $80^3$ for the full mode. Refinement is checked independent of the physical gates. Analytic $\nabla \cdot \mathbf{v} = 0$ and units are verified. A separate strict native C++ FP64 implementation checks numerical parity when an actual compiler is available; an unavailable compiler is reported as UNRESOLVED, never as a pass.

| Gate | Status | Primary metric |
|---|---|---|
| S0 | PASS | 5dd2f0d5ce2df29b |
| S1 | PASS | 0.002282591 |
| S2 | PASS | 3.509553e-11 |
| S3 | FAIL | 1.586867e-15 |
| S4 | FAIL | 2 |
| S5 | PASS | 6.709169e-12 |
| S6 | UNRESOLVED | — |
| S7 | UNRESOLVED | — |
| S8 | UNRESOLVED | — |
| S9 | UNRESOLVED | — |

Equatorial pressure-force at 0.042 m: 0.001485011 N; axial: $-0.002970023$ N. Fitted $|F_p| \propto r^{-n}$ exponents: equatorial $n = 4$, axial $n = 4$. Max relative refinement changes: $3.509553e{-}11$, $1.591789e{-}11$. A violation of the mass or slope tolerance is recorded as FAIL even when earlier independent gates failed.

## 6 Limits, identifiability and next experiment

1. Published rounded amplitude/force numbers are not independent raw force-series data or a measurement of $G$.
2. The core scale $r_c = 1.40897017 \times 10^{-15}$ m does not follow from the chosen millimetre surrogate width $a$. No matched-asymptotic core-to-macro calculation is provided.
3. The instantaneous Gaussian field is not a vortex knot, pressure-support closure, or an independently proven long-lived Euler solution.
4. Obtain a calibrated force or displacement sweep at *at least four separations*, repeat with two source materials, record mechanical transfer function and covariance, and withhold confirmation data from model selection.
5. Replace the Gaussian test provider with genuine PKLSA finite-core knots in a new registered protocol, deriving momentum transfer and test-object geometry rather than assuming $F = -V_t \nabla p$.

## References


[1] Y. Zhu, Q. Geng, B. Xue, Y. Sun, R. Zhang, X. Chen, X. Jiang, F. Azeem, M. Tierz and W. Wan (2026), "Stopped-light-enhanced gravitational force sensing," *Nature Nanotechnology*, `https://doi.org/10.1038/s41565-026-02298-8`.

[2] G. K. Batchelor (1967), *An Introduction to Fluid Dynamics*, Cambridge University Press, `https://doi.org/10.1017/CBO9780511800955`.

[3] G. W. Gibbons (2002), "The Maximum Tension Principle in General Relativity," *Foundations of Physics* **32**, 1891–1901, `https://doi.org/10.1023/A:1022370717626`.

[4] H. Cavendish (1798), "Experiments to Determine the Density of the Earth," *Philosophical Transactions of the Royal Society of London* **88**, 469–526, `https://doi.org/10.1098/rstl.1798.0022`.